METRIC
MIL-DTL-24728/1C
w/AMENDMENT 1
24 March 2023
SUPERSEDING
MIL-DTL-24728/1C
18 September 2018

DETAIL SPECIFICATION SHEET

# INTERCONNECTION BOX, FIBER OPTIC, SUBMERSIBLE, 254 X 330 MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.

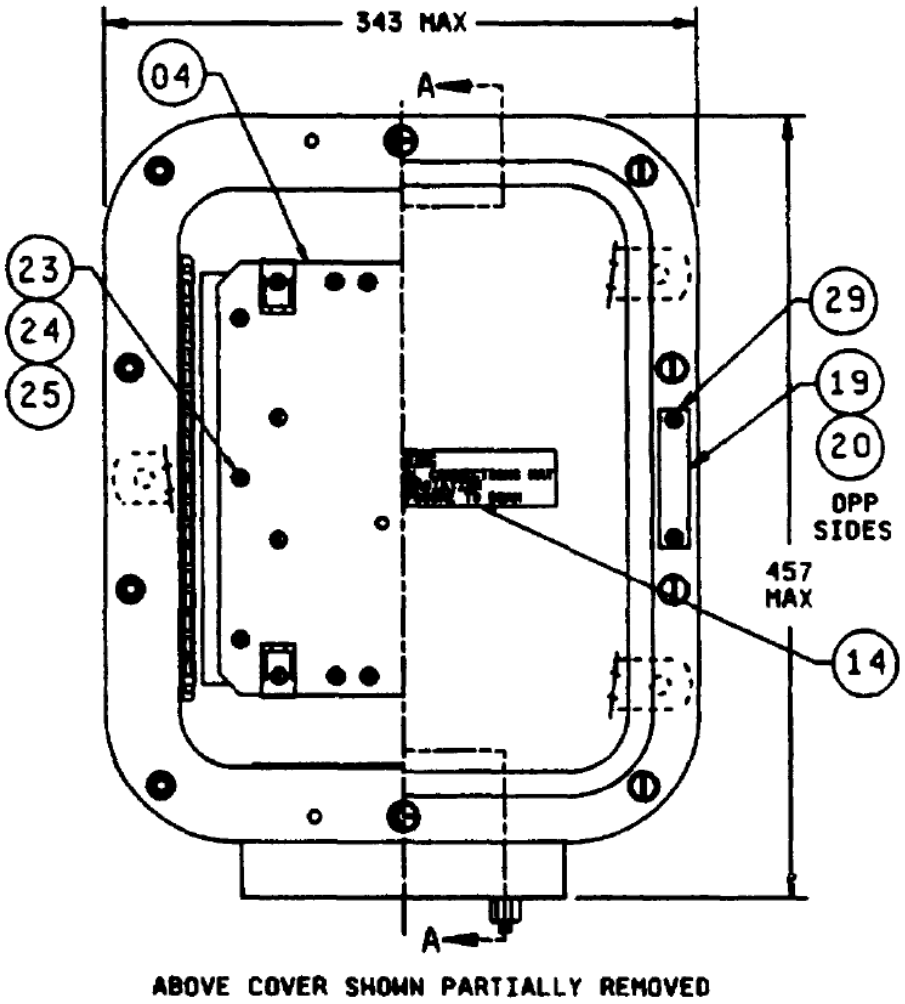



FIGURE 1. Enclosure assembly, M24728/1-005 and M24728/1-006.

AMSC N/A FSC 6099

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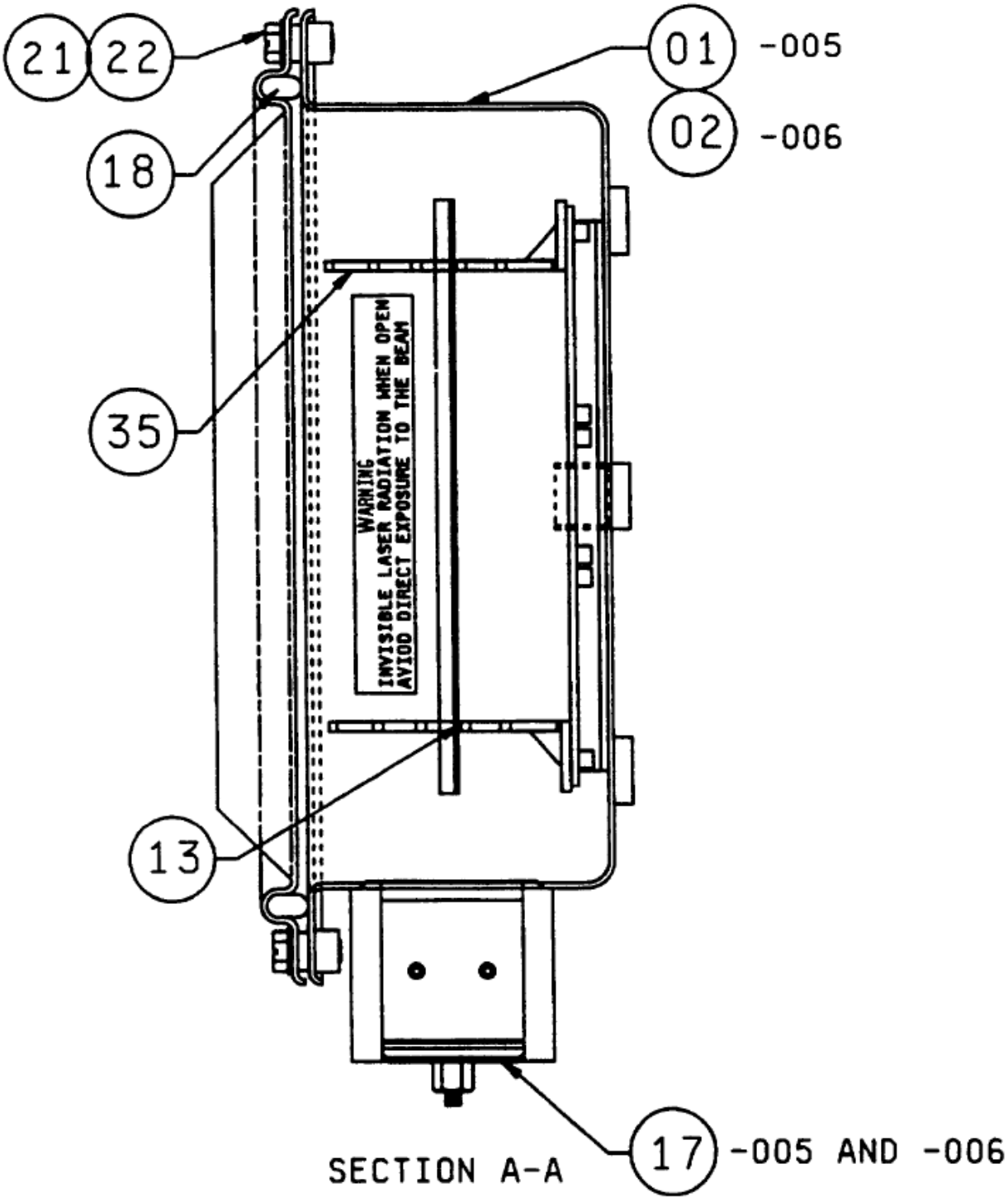
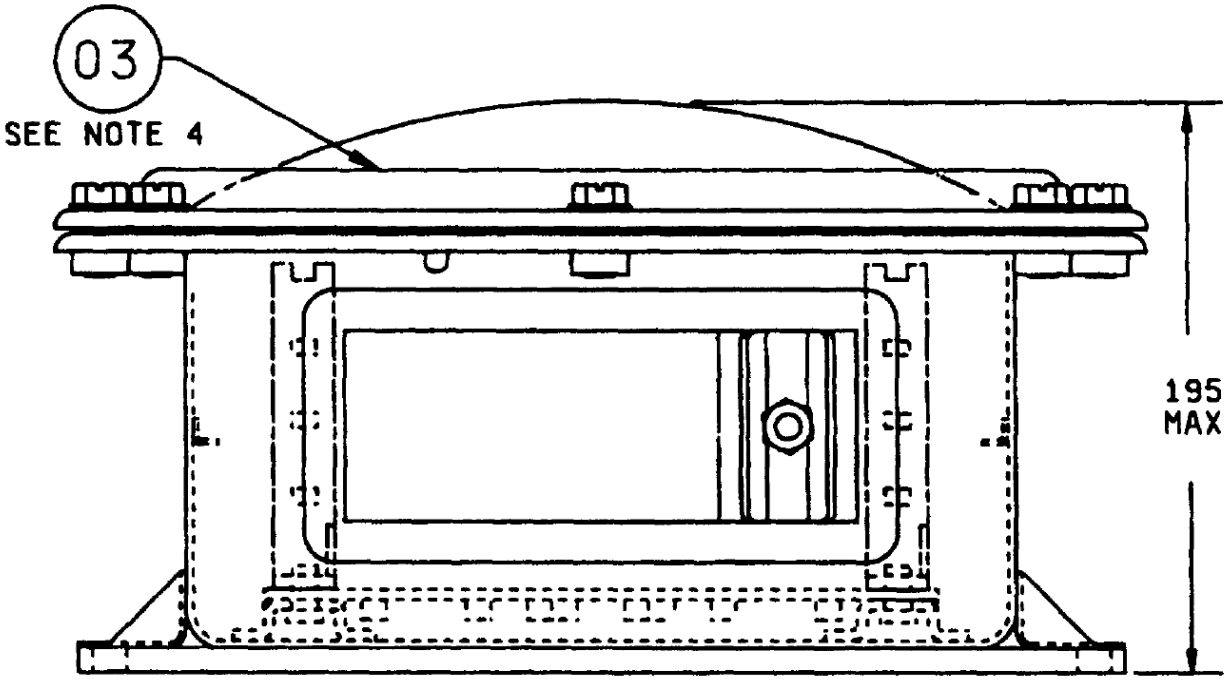



FIGURE 1. Enclosure assembly, M24728/1-005 and M24728/1-006 – Continued.



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| Millimeters | Inches |
|---|---|
| 195 | 7.7 |
| 343 | 13.5 |
| 457 | 18.0 |

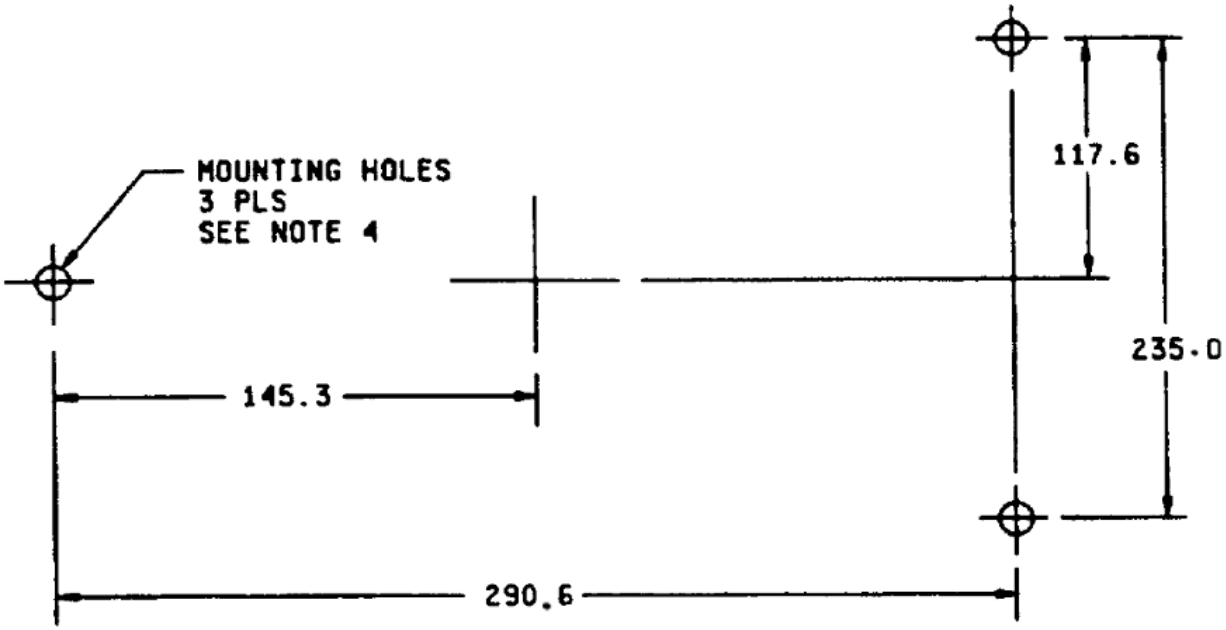
NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 1. Enclosure assembly, M24728/1-005 and M24728/1-006 – Continued.



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| Millimeters | Inches |
|---|---|
| 117.6 | 4.03 |
| 145.3 | 5.72 |
| 235.0 | 9.25 |
| 290.6 | 11.44 |

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. Dimensions are controlled by welding or brazing.
4. Mounting holes shall accommodate M12.7 x 1.95 (0.50-13) bolts.

FIGURE 2. Box mounting hole dimensions.



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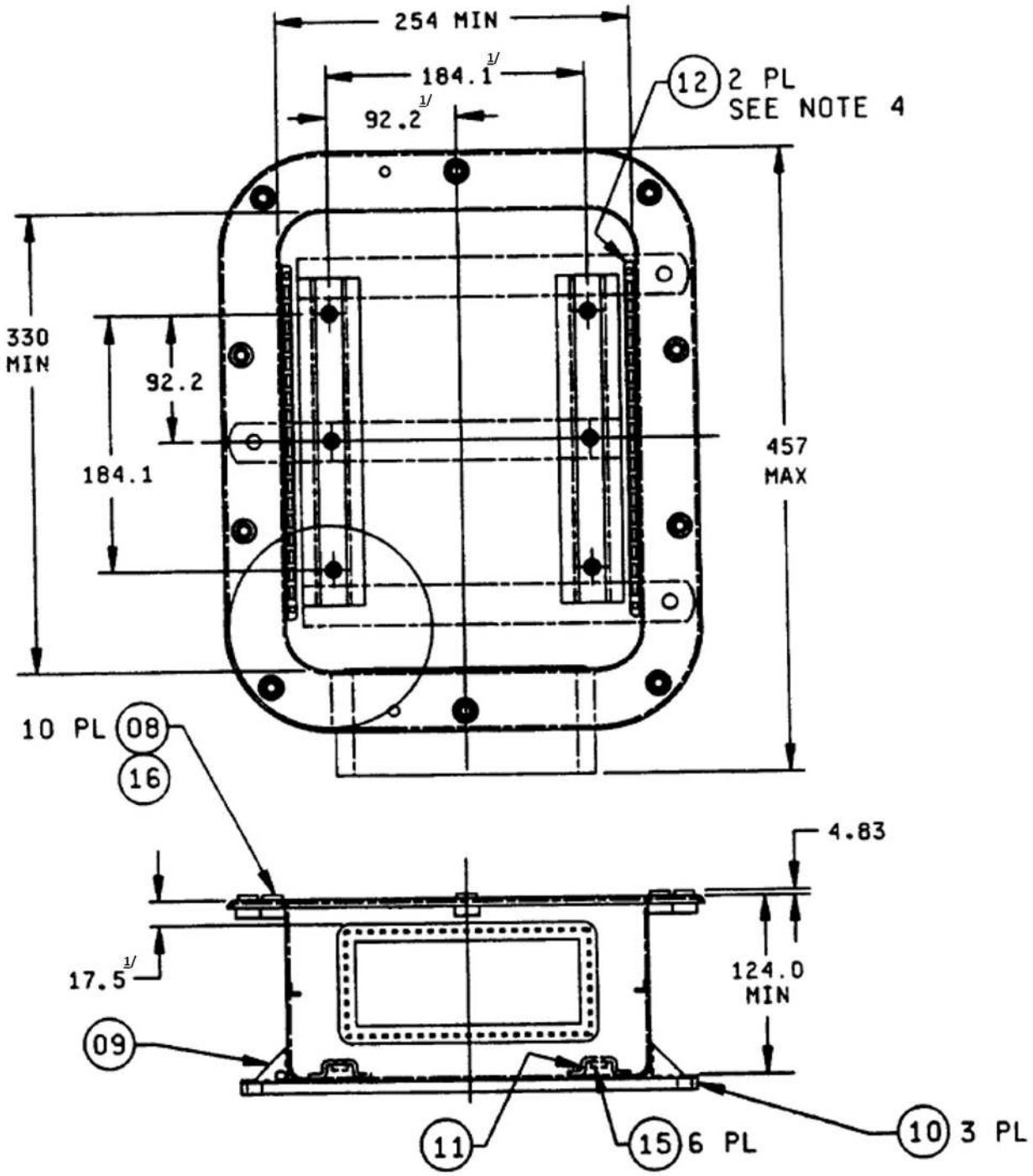
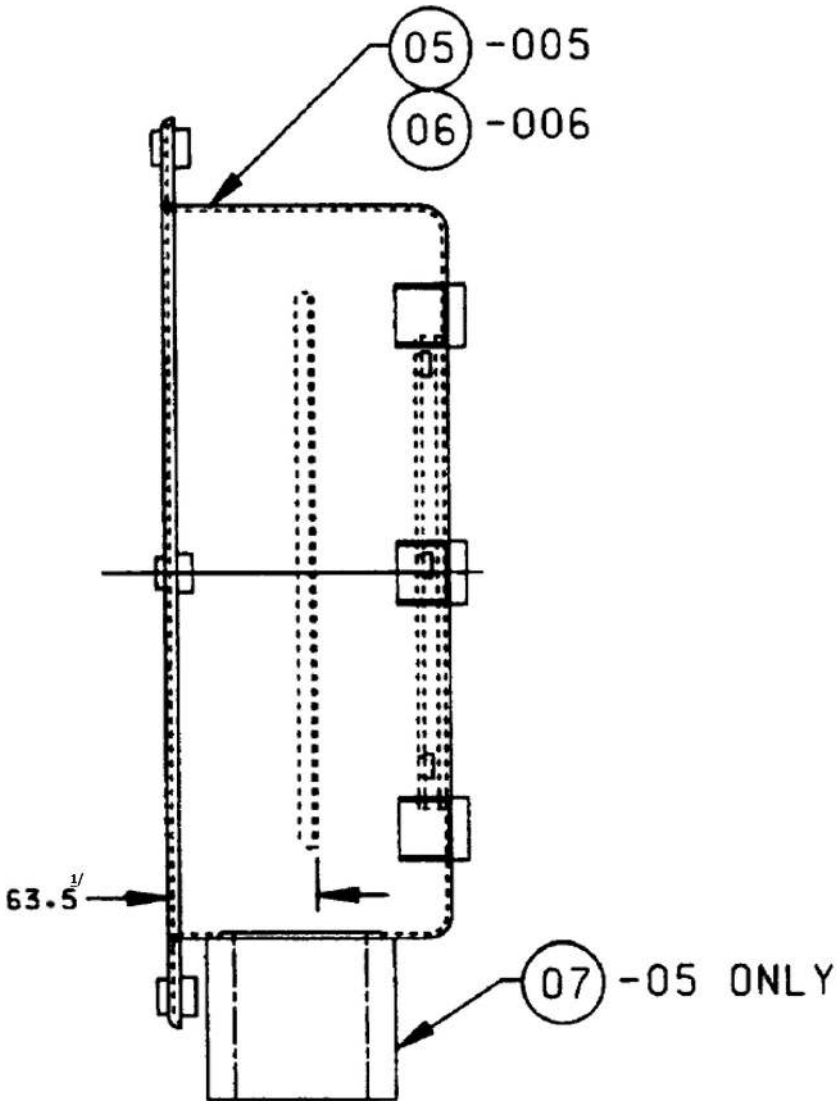



FIGURE 3. Box assembly.



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| Millimeters | Inches | Millimeters | Inches |
|---|---|---|---|
| 4.83 | 0.190 | 184.1 | 7.25 |
| 17.5 | 0.69 | 254 | 10 |
| 63.5 | 2.50 | 330 | 13 |
| 92.2 | 3.63 | 457 | 18 |
| 124.0 | 4.90 | | |

FOOTNOTE:

1/ Dimensions are controlled by welding or brazing.

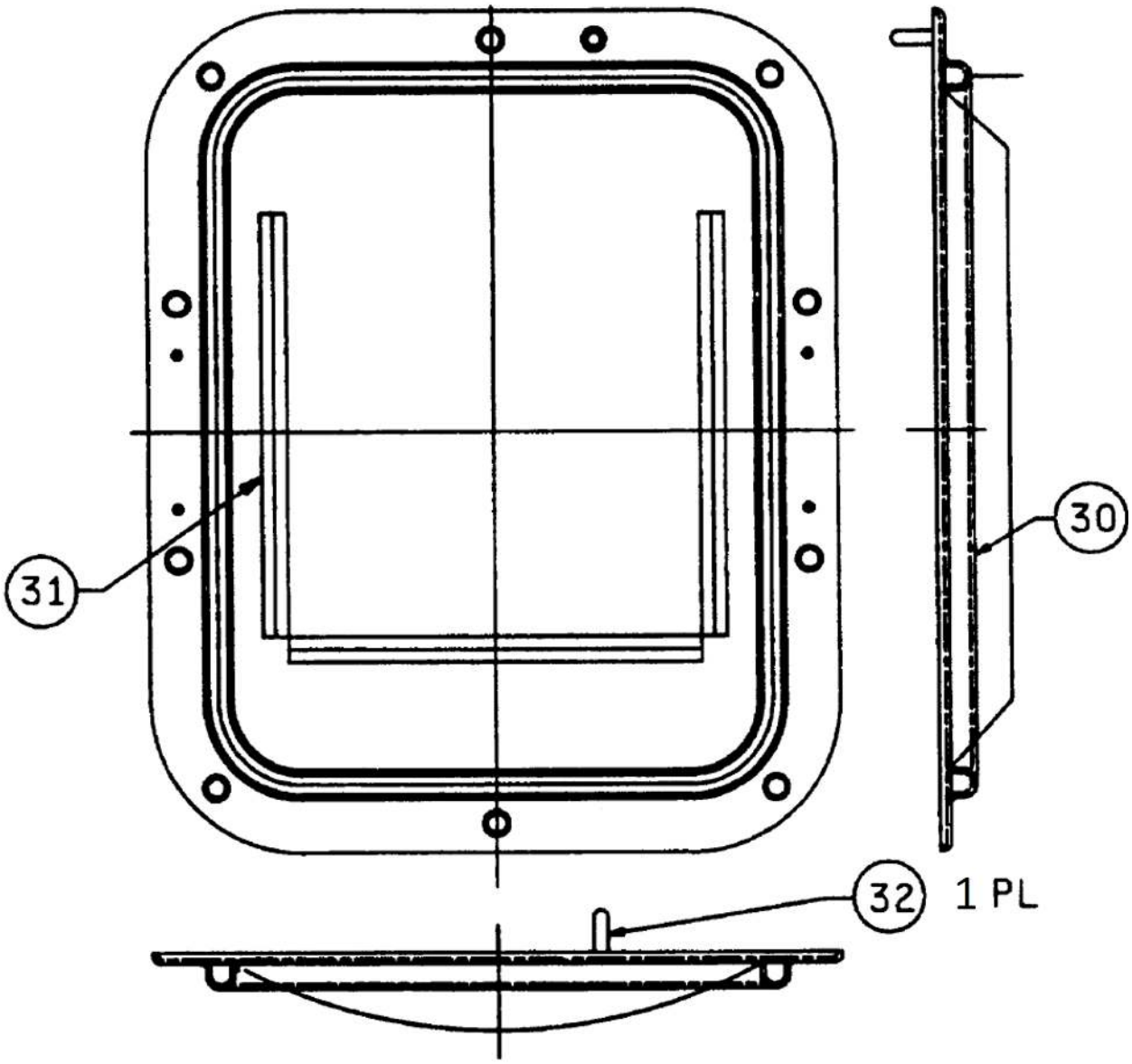
NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 3. Box assembly – Continued.



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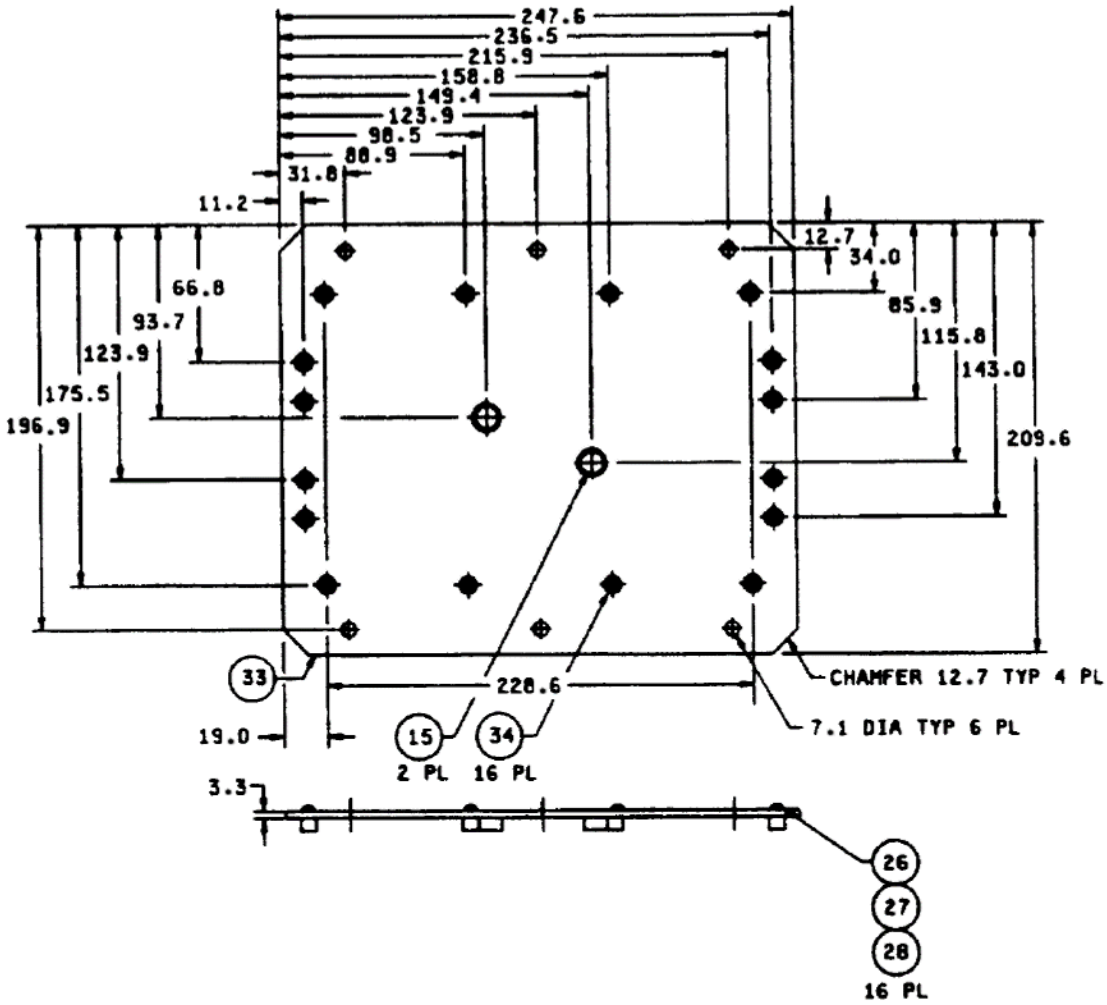

NOTES:

1. Detailed layout and placement of diagram holder is not specified.
2. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 4. Cover assembly.



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| Millimeters | Inches | Millimeters | Inches | Millimeters | Inches |
|---|---|---|---|---|---|
| 3.3 | 0.13 | 85.9 | 3.38 | 158.8 | 6.25 |
| 7.1 | 0.28 | 88.9 | 3.50 | 175.5 | 6.91 |
| 11.2 | 0.44 | 93.7 | 3.69 | 196.9 | 7.75 |
| 12.7 | 0.50 | 98.5 | 3.88 | 209.6 | 8.25 |
| 19.0 | 0.75 | 115.8 | 4.56 | 215.9 | 8.50 |
| 31.8 | 1.25 | 123.9 | 4.88 | 228.6 | 9.00 |
| 34.0 | 1.34 | 143.0 | 5.63 | 236.5 | 9.31 |
| 64.3 | 2.53 | 143.0 | 5.63 | 247.6 | 9.75 |
| 66.8 | 2.63 | 149.4 | 5.88 | | |

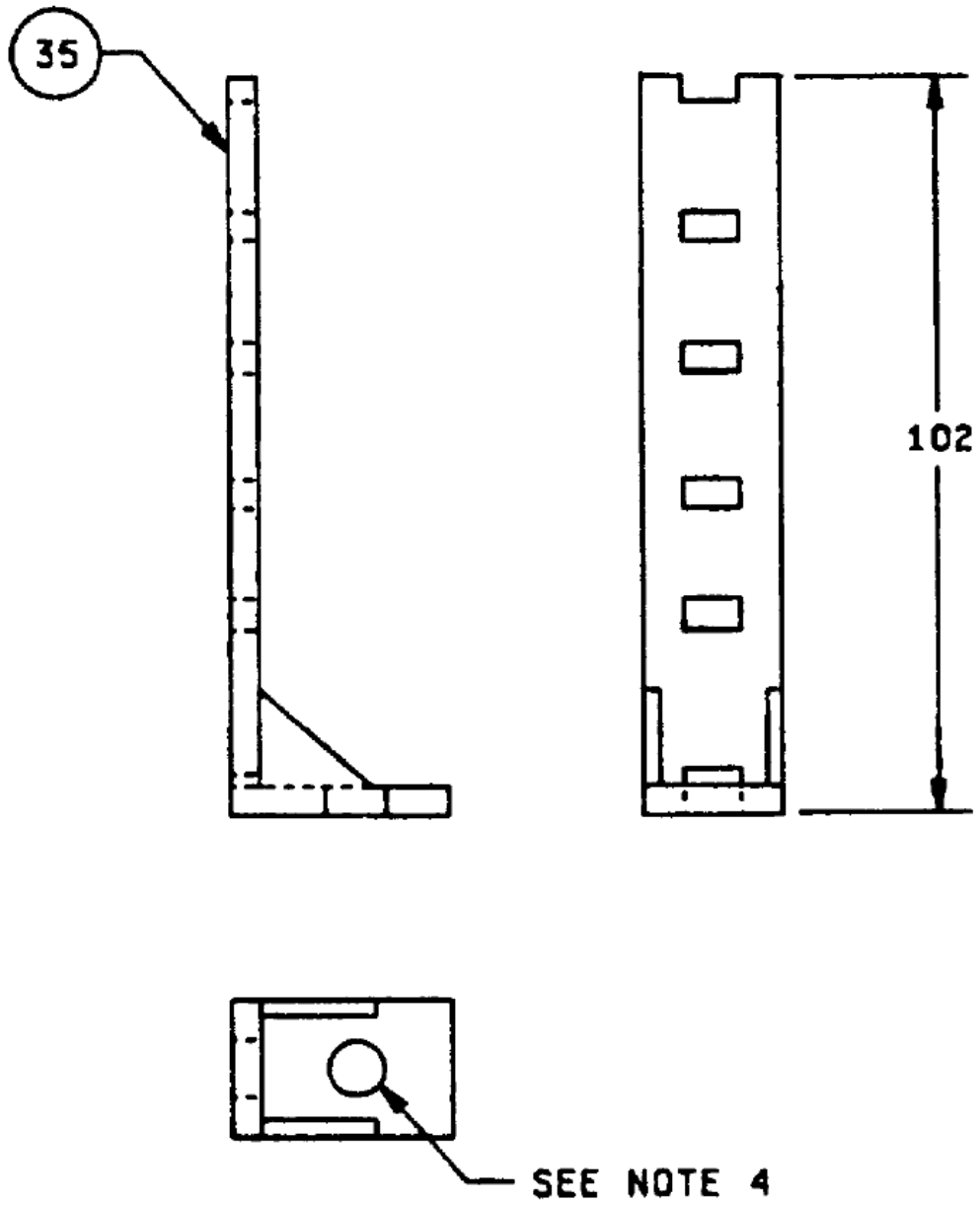
NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 5. Mounting plate assembly.



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| Millimeters | Inches |
|---|---|
| 102 | 4.02 |

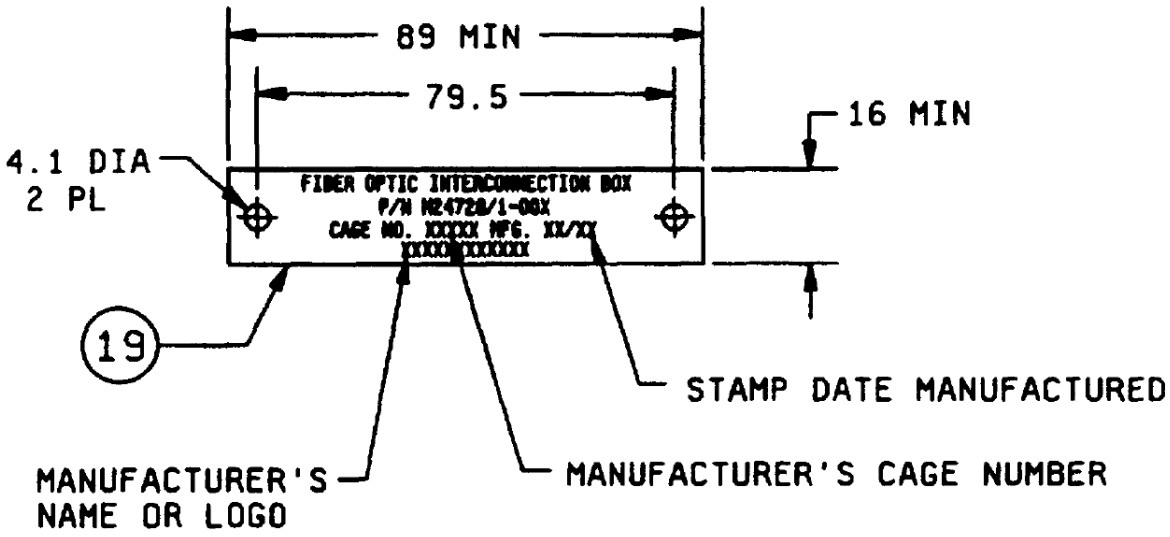
NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. Hole is 5.6 mm (0.22 inch) in diameter.
5. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 6. Tie wrap post.



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| Millimeters | Inches |
|---|---|
| 4.1 | 0.16 |
| 16.0 | 0.63 |
| 79.5 | 3.13 |
| 89.0 | 3.5 |

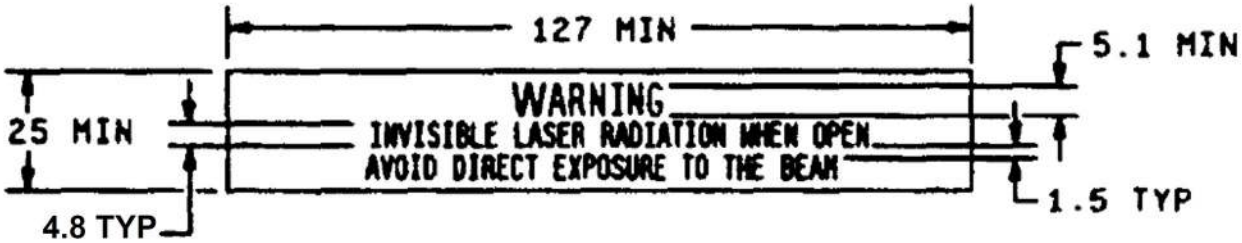
NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 7. Identification plate.



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| Millimeters | Inches |
|---|---|
| 1.5 | 0.06 |
| 4.8 | 0.19 |
| 5.1 | 0.20 |
| 25 | 1.0 |
| 127 | 5.0 |

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. DELETED.

FIGURE 8. Inside warning label.



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| Millimeters | Inches |
|---|---|
| 1.5 | 0.06 |
| 3.3 | 0.13 |
| 5.1 | 0.20 |
| 25 | 1.0 |
| 127 | 5.0 |

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. DELETED.

FIGURE 9. Outside warning label.



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TABLE I. Parts list.

| Subcomponent identification numbers, circled (see figures 1 through 9) | Part name | PIN M24728/1 dash number | | Subassembly number | | | | Remarks and reference figures |
|---|---|---|---|---|---|---|---|---|
| | | -005 | -006 | 01 | 02 | 03 | 04 | |
| 01 | Box assembly with MCT frame | 1 | | 1 | | | | 1 and 3 |
| 02 | Box assembly without MCT frame | | 1 | | 1 | | | 1 and 3 |
| 03 | Cover assembly | 1 | 1 | | | 1 | | 1 and 4 |
| 04 | Mounting plate assembly | 1 | 1 | | | | 1 | 1 and 5 |
| 05 1/, 2/ | 254 x 330 box with MCT frame | | | 1 | | | | 3 |
| 06 1/, 2/ | 254 x 330 box without MCT frame | | | | 1 | | | 3 |
| 07 3/, 4/ | MCT frame | | | 1 | | | | 3 |
| 08 2/, 3/ | Flange pad | | | 10 | 10 | | | 3 |
| 09 2/, 5/ | Gusset | | | 3 | 3 | | | 3 |
| 10 3/ | Mounting leg | | | 3 | 3 | | | 3 |
| 11 3/ | Mounting channel | | | 2 | 2 | | | 3 |
| 12 3/ | Tie wrap bracket | | | 2 | 2 | | | 3 |
| 13 6/ | Warning label (inside) | | | 1 | 1 | | | 1 and 8 |
| 14 6/ | Warning label (outside) | 1 | 1 | | | | | 1 and 9 |
| 15 7/ | M6.35 x 1.27 (0.25-20) insert | | | 6 | 6 | | 2 | 3 and 5 |
| 16 7/ | M9.53 x 1.59 (0.375-16) helicoil | | | 10 | 10 | | | 3 |
| 17 4/ | MCT wedge pack | 1 | 1 | | | | | 1 |
| 18 2/ | Gasket | 1 | 1 | | | | | 1 |
| 19 8/ | Identification plate | 1 | 1 | | | | | 1 and 7 |
| 20 8/ | Blank nameplate | 1 | 1 | | | | | 1 |
| 21 2/, 7/ | M9.53 x 1.59 (0.375-16) hex bolt | 10 | 10 | | | | | 1 |
| 22 7/ | 9.53 (0.375) flat washer | 10 | 10 | | | | | 1 |
| 23 7/ | M6.35 x 1.27 (0.25-20) x 15.9 pan head machine screws (PHMS) | 6 | 6 | | | | | 1 |
| 24 7/ | 6.35 (0.25) flat washer | 6 | 6 | | | | | 1 |
| 25 7/ | 6.35 (0.25) split lock washer | 6 | 6 | | | | | 1 |
| 26 7/ | M4.0 x 0.79 (8-32) x 12.7 PHMS | | | | | | 16 | 5 |
| 27 7/ | 4.0 (#8) flat washer | | | | | | 16 | 5 |
| 28 7/ | 4.0 (#8) split lock washer | | | | | | 16 | 5 |



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TABLE I. Parts list – Continued.

| Subcomponent identification numbers circled (see figures 1 through 9) | Part name | PIN M24728/1 dash number | | Subassembly number | | | | Remarks and reference figures |
|---|---|---|---|---|---|---|---|---|
| | | -005 | -006 | 01 | 02 | 03 | 04 | |
| 29 7/ | M3.5 x 0.79 (6-32) x 6.35 PHMS | 4 | 4 | | | | | 1 |
| 30 1/, 2/ | 254 x 330 cover | | | | | 1 | | 4 |
| 31 2/, 8/, 9/ | Diagram holder | | | | | 1 | | 4 |
| 32 2/, 8/ | Dowel pin | | | | | 1 | | 4 |
| 33 8/ | Mounting plate | | | | | | 1 | 5 |
| 34 7/ | M4.0 x 0.79 (8-32) insert | | | | | | 16 | 5 |
| 35 | Tie wrap post | 4 | 4 | | | | | 6 |

NOTES:

1/ Aluminum 3003 or 6061 in accordance with ASTM B209/B209M.

2/ See MIL-E-24142/6 for guidance.

3/ Aluminum 5052 H-32 or 6061 in accordance with ASTM B209/B209M.

4/ See MIL-DTL-24705/1 for guidance.

5/ Aluminum 5052 H-32 or 6061 in accordance with ASTM B209/B209M, 2.39 mm (0.09 inch) thick.

6/ In accordance with MIL-DTL-15024, type G, adhesive-backed, metal foil.

7/ Corrosion-resistant steel (CRES).

8/ Aluminum 5052 H-32 in accordance with ASTM B209/B209M.

9/ Constructed to hold a 216- by 279-mm (8.5- by 11.0-inch) diagram.



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REQUIREMENTS:

Design and construction. See figures 1 through 9 and table I.

Interconnection box. The interconnection box shall be constructed in accordance with M24142/6-004.

Degree of enclosure. Submersible.

MCT. The Multiple Cable Transit (MCT) shall be constructed using MIL-DTL-24705/1 as guidance. The MCT interior dimensions shall be 159.5 by 59.9 millimeters (6.28 by 2.36 inches).

Enclosure assembly construction. The enclosure assembly shall be constructed using the following combinations of materials:

a. The box, MCT frame, flange pads, gussets, mounting legs, mounting channels, tie wrap bracket, and cover shall be aluminum 6061 in accordance with ASTM B209/B209M.

b. The box and cover shall be aluminum 3003 in accordance with ASTM B209/B209M. The MCT frame, gussets, flange pads, mounting legs, mounting channels, and tie wrap bracket shall be aluminum 5052 H-32 in accordance with ASTM B209/B209M.

Enclosure assemblies constructed using aluminum 6061 shall not be welded.

Enclosure assemblies constructed using aluminum 6061 shall be dip-brazed in accordance with AWS C3.7M/C3.7.

Enclosure assemblies constructed using aluminum 6061 shall be artificially aged at 177 °C for 12 hours to a 6061-T4 condition.

Enclosure assemblies shall be class 1A conversion coated in accordance with MIL-DTL-5541 before being epoxy powder coated.

Cable interconnection interface. The cable feed-through shall be in accordance with MIL-DTL-24705 and shall accept cables having outer diameters between 6.1 and 25.4 millimeters (0.24 and 1.00 inch) constructed in accordance with MIL-PRF-85045.

Enclosure assembly stress relief. Enclosure assemblies shall be stress relieved in accordance with MIL-E-24142.

| | | |
|---|---|---|
| Mass | M24728/1-005 | 7.72 kilograms (maximum) |
| | M24728/1-006 | 5.9 kilograms (maximum) |

Interconnection box mounting. See figure 2.

Interconnect organization.

Fiber optic diagram holders. See figure 4.

Fiber optic cable tie posts. Four (see figure 6).

Fiber optic splice tray holder. One maximum (see MIL-DTL-24728/8).

Fiber optic connector module. One maximum (see MIL-DTL-24728/6).

Fiber optic splices. When outfitted with 2 MIL-DTL-24728/8 splice tray holders, the interconnection box shall organize up to 48 MIL-PRF-24623/6 fiber optic splices.

Fiber optic connector adapters. When outfitted with a MIL-DTL-24728/6 connector module, the interconnection box shall organize up to 48 MIL-DTL-83522/17 connector adapters.

Cable retention. Not applicable.

Cable seal flexing. Not applicable.



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Cable twist. Not applicable.

Compression resistance. Not applicable.

Impact. Not applicable.

Water pressure. The interconnection cover bolts and the MCT wedge pack nut shall be tightened to a minimum torque of 16.94 newton-meters (150 inch-pounds) during this test.

Flammability. Not applicable.

Salt spray. Not applicable.

Flame spread. Not applicable.

Fungus test. The fungus test shall be conducted only on polymeric materials.

Operating and storage temperature range. See temperature range 1 of MIL-DTL-24728.

Part or identifying number (PIN). See MIL-DTL-24728 and table I.

M24728/1-005, box, mounting plate, and lid, with cable entrance and wedge pack.

M24728/1-006, box, mounting plate, and lid, without cable entrance.

M24728/1-001, M24728/1-002, M24728/1-003, and M24728/1-004 have been canceled without replacement.

Gasket size. The gasket size shall be 263.53 millimeters (10.375 inches) wide by 339.73 millimeters (13.375 inches) long. This corresponds to PIN M24142/6-004 of MIL-E-24142/6.

Insert block. Insert blocks compatible with the MCT frame are specified in MIL-DTL-24705/1.

Qualification by similarity. Manufacturers who produce products for both MIL-DTL-24728/2 and this specification sheet who are qualified under MIL-DTL-24728/2, and who pass the water pressure test and size inspection specified herein, are qualified under this specification sheet.

Measurement system. The U.S. Government preferred system of measurement is the metric (SI) system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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CONCLUDING MATERIAL

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Navy – SH
Air Force – 85
DLA – CC

Review activities:
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Navy – AS, CG, EC, MC, YD
Air Force – 19
DLA – GS

Civil agency:
NASA – NA

Preparing activity:
Navy – SH
(Project 6099-2022-005)

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